

Comprehensive Deployment & Database Migration Plan

Development → production deploy server · Local MySQL → Railway cloud database · Verification, rollback, and disaster recovery

DOCUMENT PURPOSE

End-to-end operational guide for ZAIZENS HMS upgrades, external server deployment, and MySQL replication to Railway

DEVELOPMENT ROOT

C:\HMS_JS

UPDATE PACKAGE

C:\HMS_JS\Update

PRODUCTION INSTALL

C:\Program Files\ZAIZENS\HMS

DATABASE SYNC

docs/scripts/ · Railway MySQL

Version

3.0 — June 2026

Related guides

Local Deployment · MySQL→Railway · License Server Railway

Audience

Implementation team · IT operations · Vendor

Classification

Confidential — client operations

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1. Executive overview

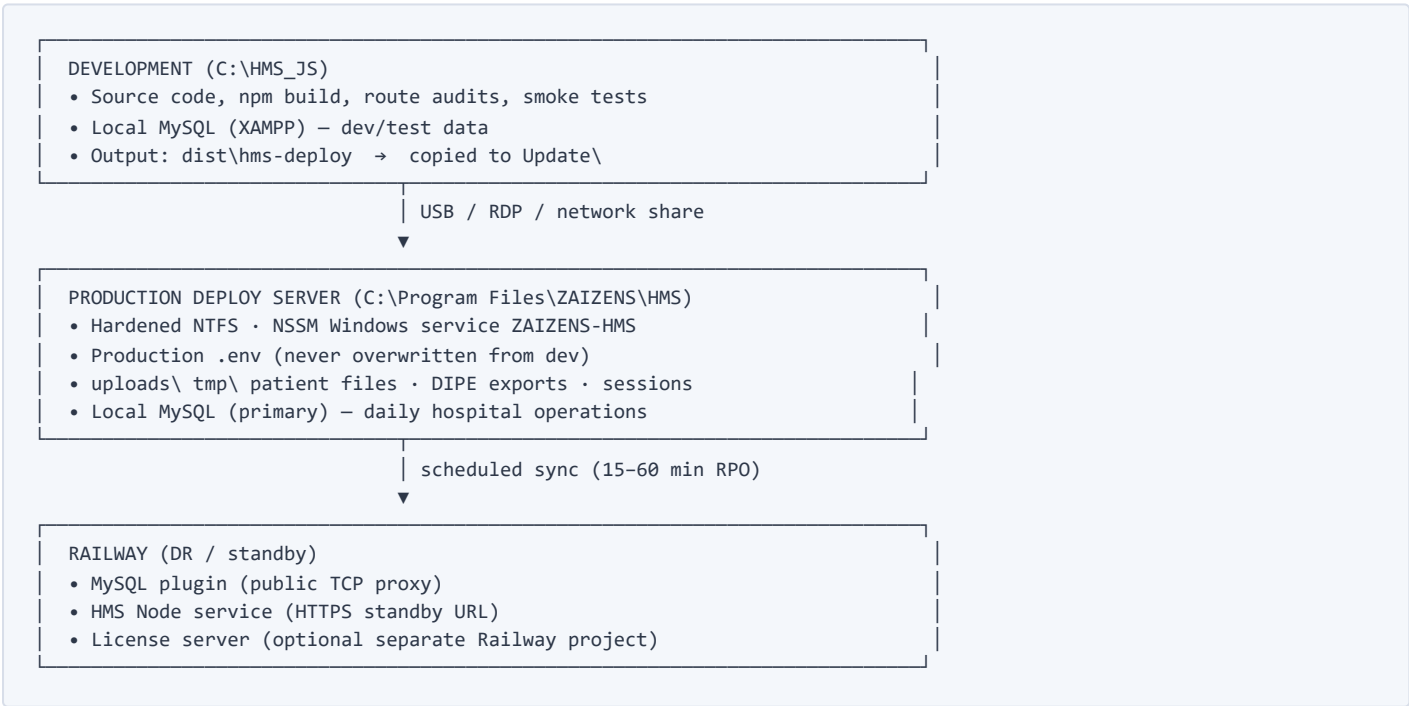
Goal: Safely promote tested HMS application code from the development workspace to a hardened external Windows server, while maintaining a warm copy of the hospital database on Railway for disaster recovery and optional cloud workflows.

This plan covers three coordinated workstreams:

Workstream	Scope	Primary artifacts
Application deploy	Dev build → Update folder → production Program Files install	<code>build-update-package.ps1</code> , <code>apply-deploy-update.ps1</code>
Database migration	Local MySQL (XAMPP / Windows) → Railway MySQL	<code>sync-mysql-to-railway.js</code> , <code>bidirectional-sync-local-railway.js</code>
Operations	Backups, service install, NTFS hardening, restore	<code>docs/scripts/04-12</code> , nightly backup task

Single writer rule: Only one environment should accept clinical writes at a time. Scheduled sync is warm standby (local → Railway), not active-active. Use bidirectional sync only with explicit table scope and change-control.

2. Architecture & environments



2.1 Environment matrix

Component	Development	Production server	Railway standby
App root	C:\HMS_JS	C:\Program Files\ZAIZENS\HMS	Railway service root

Component	Development	Production server	Railway standby
Database	Local MySQL	Local MySQL (primary)	Railway MySQL
<code>.env</code>	Dev credentials	Production — preserved on deploy	Railway variables
UI bundles	<code>public/dist/*.js</code>	Deployed via Update	Same build as production
User URL	<code>localhost:3000</code>	Hospital LAN hostname	HTTPS Railway domain (DR)

3. Pre-deployment checklist

PHASE 0 — BEFORE ANY PRODUCTION CHANGE

- ☐ All changes tested on dev: login, OPD queue, billing, pharmacy, statutory reports
- ☐ Route audit clean: `node scripts/audit-routes.js`
- ☐ Smoke routes pass: `node scripts/smoke-routes.js`
- ☐ UI rebuilt: `npm run build:ui` produces fresh `hms-ui.js` (~1.5–2.2 MB)
- ☐ Production backup verified (DB dump + file backup within 24 h)
- ☐ Deploy window communicated to clinical staff
- ☐ Administrator PowerShell available on deploy server
- ☐ Update folder copied and verified (see Section 5)
- ☐ Railway sync env configured if DB migration is in scope
- ☐ License keys and `LICENSE_SERVER_URL` unchanged unless planned

4. Build update package (development machine)

PHASE 1 — DEV WORKSPACE

Run on `C:\HMS_JS` after merging and testing all changes:

```
powershell -ExecutionPolicy Bypass -File C:\HMS_JS\scripts\build-update-package.ps1
```

This script:

1. Runs `npm run build:ui` (React bundles: `hms-ui.js`, `hms-login.js`, `hms-print.js`)
2. Runs `node scripts/build-deploy-package.js` → `dist\hms-deploy`
3. Robocopies deploy output to `C:\HMS_JS\Update` (replaces previous Update folder)
4. Writes `UPDATE-README.txt` with timestamp and verify steps

Critical: Deploying backend files without `public\dist\hms-ui.js` leaves users on an old UI. Always confirm bundle size and date before copying to production.

4.1 Optional dev data export

For catalog/pharmacy seed on production (only when intentional):

```
node scripts/export-pharmacy-deploy-data.js
# Produces Update\data\deploy-data-export.json
```

Apply on server with `-ImportDeployData` flag (Section 6).

5. Transfer to deploy server

PHASE 2 — PHYSICAL OR REMOTE COPY

Copy the **entire** `C:\HMS_JS\Update` folder to the deploy server (same path recommended):

- USB drive, RDP drag-and-drop, or network share
- Include `scripts\apply-deploy-update.ps1`, all routes, lib, views, `public\dist\`

5.1 Verify after copy

- ☐ `C:\HMS_JS\Update\app.js` exists
- ☐ `C:\HMS_JS\Update\public\dist\hms-ui.js` exists (~1500–2200 KB)
- ☐ `C:\HMS_JS\Update\scripts\apply-deploy-update.ps1` present
- ☐ New modules present (e.g. `routes\statutoryReports.js`, `lib\erOrderAutoCharge.js`)
- ☐ `MANUAL-COPY-README.txt` reviewed

Never overwrite on deploy server: production `.env`, `uploads\`, `tmp\`, patient attachments, or DIPE export data.

6. Apply production update

PHASE 3 — DEPLOY SERVER (ADMINISTRATOR)

6.1 Automated apply (recommended)

```
powershell -ExecutionPolicy Bypass -File C:\HMS_JS\Update\scripts\apply-deploy-update.ps1
```

The script performs:

- 1 Stop HMS** — NSSM service `ZAIZENS-HMS` or running `node.exe` processes
- 2 Backup** — robocopy to `deploy-backups\HMS-backup-YYYYMMDD-HHmm` (with `/B` for backup mode on hardened trees)
- 3 ACL unlock** — grants Administrators modify on install tree (required after NTFS hardening)
- 4 Apply files** — robocopy Update → `C:\Program Files\ZAIZENS\HMS`, excluding `.env`
- 5 Verify UI bundle** — confirms `public\dist\hms-ui.js` present
- 6 Optional data import** — `-ImportDeployData` or `-PullPharmacyFromRailway`
- 7 Start HMS** — NSSM start or background `node app.js`

6.2 Script flags

Flag	Effect
-SkipBackup	Skip pre-deploy robocopy backup (use only if recent backup exists)
-SkipServiceRestart	Do not stop/start service (manual control)
-ImportDeployData	Import dev catalog export from Update\data\
-PullPharmacyFromRailway	Pull pharmacy tables from Railway to local
-HmsRoot	Override install path (default: Program Files\ZAIZENS\HMS)
-UpdateSource	Override Update folder path

6.3 Manual robocopy (fallback)

```
nssm stop ZAIZENS-HMS
robocopy "C:\Program Files\ZAIZENS\HMS" "D:\HMS-Data\deploy-backups\HMS-backup-YYYYMMDD" /E /B
robocopy C:\HMS_JS\Update "C:\Program Files\ZAIZENS\HMS" /E /B /IS /IT /XF .env
nssm start ZAIZENS-HMS
```

6.4 First-time production install

If no existing install, use the implementation kit under docs\scripts\ :

- 1 04-setup-data-folders.ps1 — uploads, tmp, logs
- 2 05-install-env.ps1 — copy and edit 02-tssf-soa.env
- 3 01-create-mysql-database.sql — create schema
- 4 07-install-hms-service.ps1 — register NSSM service
- 5 06-apply-ntfs-hardening.ps1 — restrict filesystem ACLs
- 6 09-register-backup-task.ps1 — nightly backup schedule

7. Local MySQL → Railway migration

PHASE 4 — DATABASE REPLICATION

Warm standby sync keeps Railway MySQL aligned with the on-premises primary for disaster recovery. This is *not* a one-time cutover unless you explicitly fail over (Section 11).

7.1 Strategy summary

Mode	Script	Use case
One-way full refresh	scripts/sync-mysql-to-railway.js	Initial Railway seed, scheduled DR sync
Table-scoped sync	docs/scripts/sync-local-railway.js	Directional sync of named tables
Replicate (local → Railway)	docs/scripts/replicate-local-to-railway.js	Incremental table replication
Bidirectional	docs/scripts/bidirectional-sync-local-railway.js	Controlled two-way sync (advanced)

7.2 Railway MySQL setup

- 1 Railway project → **New** → **Database** → **MySQL**
- 2 Note variables: MYSQLHOST , MYSQLPORT , MYSQLUSER , MYSQLPASSWORD , MYSQLDATABASE
- 3 Enable **public networking** on MySQL plugin (required for sync from hospital WAN)
- 4 Use strong passwords; restrict source IP if Railway plan supports allowlisting

7.3 Configure sync environment

Copy example and edit (do not commit secrets):

```
copy docs\scripts\railway-sync.env.example docs\scripts\railway-sync.env

# Local primary (XAMPP or Windows MySQL)
LOCAL_DB_HOST=localhost
LOCAL_DB_PORT=3306
LOCAL_DB_USER=tssfsoas_Admin
LOCAL_DB_PASSWORD=***
LOCAL_DB_NAME=tssfsoas_HMS
XAMPP_MYSQL_BIN=C:\xampp\mysql\bin

# Railway MySQL (from plugin)
RAILWAY_MYSQL_HOST=crossover.proxy.rlwy.net
RAILWAY_MYSQL_PORT=36338
RAILWAY_MYSQL_USER=root
RAILWAY_MYSQL_PASSWORD=***
RAILWAY_MYSQL_DATABASE=railway

SYNC_DUMP_PATH=C:\HMS-Backups\railway-sync\latest.sql
```

Alternatively, place db-sync.env in repo root or scripts\ . The sync core loads HMS .env DB_* when LOCAL_DB_USE_HMS_ENV is not disabled.

7.4 Initial migration (full dump)

```
cd C:\HMS_JS
powershell -ExecutionPolicy Bypass -File docs\scripts\run-railway-sync.ps1
```

Or directly:

```
node scripts/sync-mysql-to-railway.js
```

Process:

- 1. `mysqldump --single-transaction` from local MySQL
- 2. Connect to Railway MySQL over TLS
- 3. Drop and re-import tables (full refresh)
- 4. Verify table counts and sample row checks

Large databases (multi-GB) may take several minutes. Schedule during low activity. Railway HMS should remain in maintenance or unused until sync completes.

7.5 Railway HMS standby app

Deploy HMS as a Railway Node service linked to the same MySQL. Minimum environment:

Variable	Value
<code>DB_HOST</code> / <code>DB_PORT</code>	Railway MySQL host and port
<code>DB_USER</code> / <code>DB_PASSWORD</code> / <code>DB_NAME</code>	From MySQL plugin
<code>SESSION_SECRET</code>	Match local or accept re-login on failover
<code>LICENSE_SERVER_URL</code>	Vendor license server HTTPS URL
<code>HMS_PUBLIC_URL</code>	Railway HMS public URL (webhooks)

8. Bidirectional & scheduled sync

8.1 Scheduled one-way sync (recommended DR)

Register Windows Task Scheduler to run every 15–60 minutes:

```
powershell -ExecutionPolicy Bypass -File docs\scripts\register-replicate-scheduled-task.ps1
```

Logs: `C:\HMS-Backups\railway-sync\logs\` Or `tmp\railway-sync-logs\`

8.2 Bidirectional sync (controlled)

```
powershell -ExecutionPolicy Bypass -File scripts\run-bidirectional-sync.ps1
powershell -ExecutionPolicy Bypass -File scripts\run-bidirectional-sync.ps1 -DryRun
powershell -ExecutionPolicy Bypass -File scripts\run-bidirectional-sync.ps1 -Tables
tbl_service_catalog,tbl_inventory_item
```


Split-brain risk: Bidirectional sync can overwrite production data. Use table filters, dry-run first, and change-control.
Normal operations: local primary → Railway only.

8.3 RPO / RTO targets

Metric	Target	Monitor
RPO	≤ sync interval (e.g. 15 min)	Last successful log timestamp
RTO	≤ 30 min user redirect	Quarterly DR drill
Sync success	Every scheduled run	Task Scheduler history
Railway health	Weekly	GET /__health on standby URL

9. Post-deploy verification

PHASE 5 — GO-LIVE CHECKS

- ☐ HMS service running: `nssm status ZAIZENS-HMS` or process on port 3000
- ☐ Health endpoint: `GET /__health` — route mounts OK
- ☐ Hard refresh browser: Ctrl+F5 (clear `hms-ui.js` cache)
- ☐ Login as admin and ward user
- ☐ OPD queue and consultation flow
- ☐ Pharmacy → Products and Dispensing tabs
- ☐ Statutory reports: `/tax/statutory-reports`
- ☐ Emergency: place order → charge on credit tab (ER auto-charge)
- ☐ Print slip / receipt generation
- ☐ License heartbeat (Solution Subscriptions UI)
- ☐ If sync enabled: verify Railway row counts match local sample tables

```
node docs\scripts\12-run-diagnostic.ps1
node scripts\smoke-routes.js
```

10. Rollback procedures

10.1 Application rollback

- 1 Stop HMS service
- 2 Restore from deploy backup: `deploy-backups\HMS-backup-*`

```
robocopy "D:\HMS-Data\deploy-backups\HMS-backup-YYYYMMDD-HHmm" "C:\Program Files\ZAIZENS\HMS" /E /B /IS /IT
```

- 3 Confirm production `.env` intact
- 4 Start service; hard refresh clients

10.2 Database rollback

```
powershell -ExecutionPolicy Bypass -File docs\scripts\11-restore-database.ps1 -DumpFile "D:\HMS-Data\mysql-backups\tssfsoas_HMS_YYYYMMDD.sql"
```

Requires HMS stopped; restores from nightly mysqldump. Document any data entered after backup time as lost.

11. Disaster recovery & failover

Stop writes on primary first. If primary is dead, run final sync from readable backup media if possible, then activate Railway.

- 1 Confirm primary failure (hardware, MySQL, network)
- 2 Final sync if local MySQL readable: `node scripts/sync-mysql-to-railway.js`
- 3 Verify Railway DB — patient row counts vs last known good
- 4 Start Railway HMS; test `/__health`
- 5 Redirect users: DNS, emergency URL broadcast, or proxy rules
- 6 Update `HMS_PUBLIC_URL` on license server for push webhooks
- 7 Document data loss window (time since last sync)

11.1 Fail-back to local

1. Repair local server; restore OS and MySQL from backup
2. Reverse sync: dump Railway → import local (swap LOCAL/RAILWAY in env), or restore local backup + manual gap fill
3. Full UAT on local before switching users back
4. Re-enable scheduled local → Railway sync

12. Security & NTFS hardening

Production installs under `Program Files` use restricted ACLs (`06-apply-ntfs-hardening.ps1`). Deploy scripts temporarily grant Administrators modify access during updates.

- Run deploy as **Administrator** — non-elevated runs produce ACL noise and partial copies
- Never deploy private license signing keys to hospital servers
- Keep `railway-sync.env` and MySQL credentials out of git
- Do not expose XAMPP MySQL to `0.0.0.0` without firewall IP restrictions
- Rotate MySQL passwords via `01b-rotate-to-hardened-password.sql` after initial install

13. Troubleshooting

Symptom	Cause / fix
Thousands of "Access is denied" during deploy	NTFS hardening + non-elevated icacs; run PowerShell as Administrator; fixed icacs grant syntax in apply-deploy-update.ps1
Robocopy exit 9 on backup	Files locked by node.exe; script stops processes first; use <code>/B</code> backup mode; retry or <code>-SkipBackup</code> if recent backup exists
Old UI after deploy	Missing or stale <code>hms-ui.js</code> ; rebuild Update package; hard refresh Ctrl+F5
<code>ZAIZENS-HMS</code> service not found	Run <code>07-install-hms-service.ps1</code> ; or start manually with <code>node app.js</code>
Railway <code>ECONNREFUSED</code>	Enable public networking; verify host/port/password
<code>mysqldump</code> not found	Set <code>XAMPP_MYSQL_BIN</code> or install MySQL client
Import timeout on Railway	Run off-peak; increase Node memory; exclude log tables if needed
Schema errors on import	Ensure HMS versions match; script sanitizes MariaDB→MySQL8 defaults
404 on new routes	Run route audit; verify route file deployed; check <code>/__health</code> mount list

Appendix A — Scripts & reference

A.1 Development & deploy

Script	Purpose
<code>scripts/build-update-package.ps1</code>	Build Update folder from dev
<code>scripts/apply-deploy-update.ps1</code>	Apply Update on production server
<code>Update/MANUAL-COPY-README.txt</code>	Manual copy checklist
<code>Update/DEPLOY-README.txt</code>	First-time deploy notes

A.2 Database sync

Script	Purpose
<code>scripts/sync-mysql-to-railway.js</code>	Full local → Railway dump/import
<code>docs/scripts/railway-sync-core.js</code>	Shared sync utilities
<code>docs/scripts/sync-local-railway.js</code>	Directional table sync
<code>docs/scripts/replicate-local-to-railway.js</code>	Scheduled replication
<code>docs/scripts/bidirectional-sync-local-railway.js</code>	Two-way sync with filters
<code>docs/scripts/run-railway-sync.ps1</code>	Wrapper with env load

Script	Purpose
scripts/run-bidirectional-sync.ps1	Bidirectional wrapper

A.3 Implementation kit (docs/scripts/)

04-setup-data-folders.ps1	08-tssf-hms-backup.ps1
05-install-env.ps1	09-register-backup-task.ps1
06-apply-ntfs-hardening.ps1	10-integrity-baseline.ps1
06b-install-nssm.ps1	11-restore-database.ps1
07-install-hms-service.ps1	12-run-diagnostic.ps1

A.4 Related PDF guides

- docs/LOCAL-DEPLOYMENT-IMPLEMENTATION.pdf — on-premises install & security
- docs/MYSQL-REPLICATION-LOCAL-TO-RAILWAY.pdf — DR sync detail
- docs/LICENSE-SERVER-RAILWAY-DEPLOYMENT.pdf — vendor license server
- docs/WINDOWS-LOCAL-BACKUP-DR.html — backup strategy

A.5 Document control

Version	Date	Changes
3.0	June 2026	Unified deployment + MySQL→Railway migration plan; premium/professional layout; apply-deploy ACL fixes; bidirectional sync; statutory/ER upgrade notes